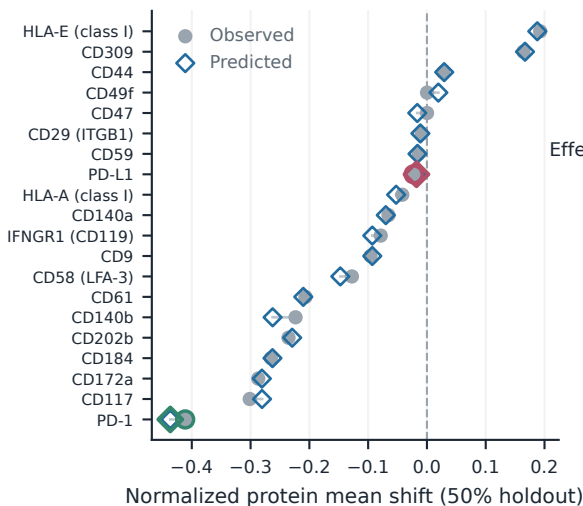
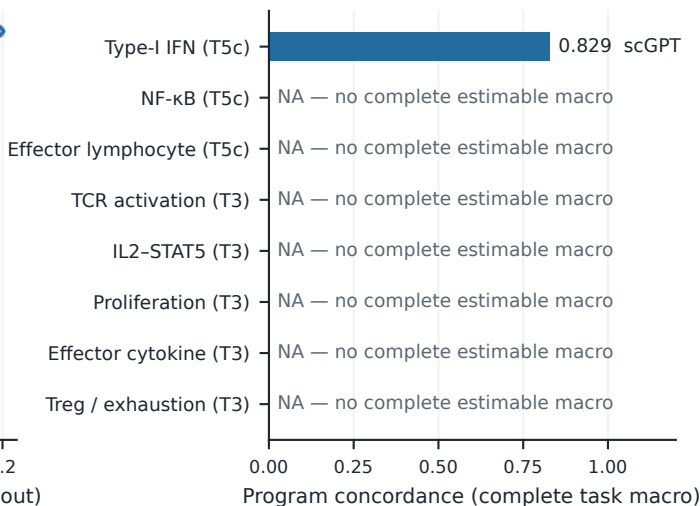


Immune readouts retain distinct marker, program and lineage information

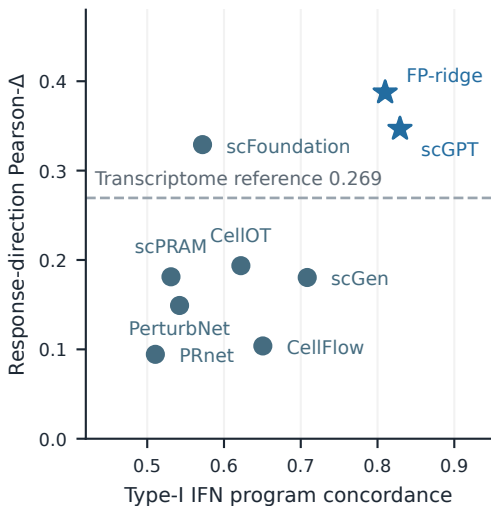
a Surface-marker shifts



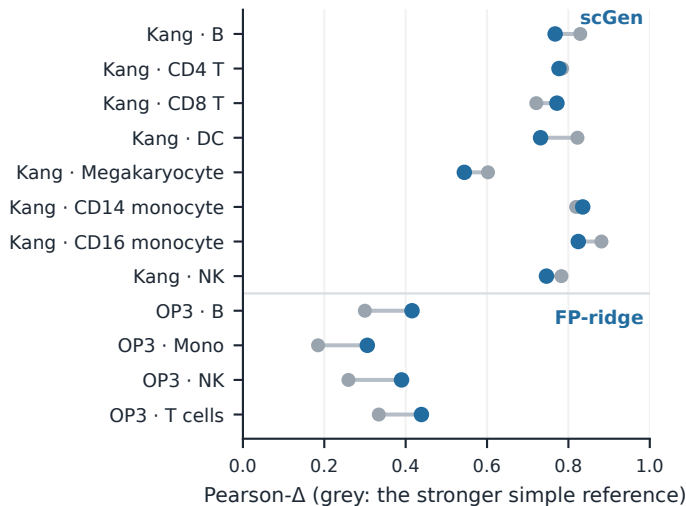
b Immune-program concordance



c Program and transcriptome agreement



d Prediction varies by lineage



Protein reference is fitted within modality. Undefined program correlations are not zero recovery. Local point advantages do not establish task-wide or multiplicity-adjusted support.